BIO 004 · SOLANO COMMUNITY COLLEGE · FALL 2026

Human Anatomy.

Competency Packet · Fall 2026

5.0 units · 17 weeks · August 17 to December 18, 2026

Vacaville Center · Lecture VC 118 · Lab VC 1137

C-ID BIOL 110B · Cal-GETC Area 5B with laboratory · Transferable to CSU and UC

For students A checklist of everything this course asks you to be able to do. Work down it. When you can do the thing without looking, that one is finished

For evaluating institutions The full assessed content of BIO 004, by module and topic, each competency tagged for written lecture examination, laboratory practical, or both

Also in Canvas The course syllabus and the Course Policies and Articulation Reference

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Course Competencies

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These 193 competencies are the whole of what BIO 004 asks you to be able to do. They sit under the three approved course outcomes, they are what the Mastery OS tracks, and they are what the five lecture exams and five lab practicals are written from. Nothing is assessed that is not on this list.

How to read this list. Competencies are grouped first by module, then by topic within the module, in teaching order. Each one is tagged for where you are held to it: LECTURE means it is on the lecture exam for that module, and LAB means you have to produce it in the lab practical, on a slide, a model, a chart, or prosected material. Many carry both, because in anatomy naming a structure and finding it are two different skills.

COMPETENCY LOAD BY MODULE

MOD	MODULE	TOTAL	LECTURE	LAB	CORE
1	Foundations to Integumentary	23	23	13	13
2	Skeletal System and Joints	30	30	28	14
3	Cardiovascular, Muscle and Blood	42	32	33	23
4	Respiratory, Lymphatic, Digestive and Endocrine	36	34	28	17
5	Urinary, Reproductive and the Nervous System	62	61	51	35
	All five modules	**193**	**180**	**153**	**102**

Module 1. Foundations to Integumentary

23 COMPETENCIES · ASSESSED ON EXAM 1

FOUNDATIONS

1. **Levels of structural organization.** List the six levels of structural organization in order from chemical to organismal and state what each level contains. LECTURE
2. **Anatomical position.** Place the body in anatomical position, define supine and prone, and explain why anatomical position is the fixed reference for every directional term. LECTURE
3. **Anatomical planes and sections.** Name the sagittal, midsagittal, parasagittal, frontal, transverse, and oblique planes and state the two parts each divides the body into. LECTURE LAB
4. **Directional terms.** Use the paired directional terms precisely, including superior and inferior, anterior and posterior, medial and lateral, proximal and distal, superficial and deep, cranial and caudal, and ipsilateral and contralateral. LECTURE
5. **Regional terms.** Identify the major anterior and posterior regional terms and sort each as axial or appendicular. LECTURE LAB

BODY CAVITIES AND REGIONS

6. **Body cavities.** Distinguish the dorsal and ventral body cavities and name every subdivision of each, including the cranial, vertebral, thoracic, pleural, pericardial, abdominal, and pelvic cavities. LECTURE LAB
7. **Serous membranes.** Define a serous membrane and differentiate its parietal and visceral layers across the pleura, pericardium, and peritoneum. LECTURE LAB
8. **Mediastinum.** Map the boundaries and the four compartments of the mediastinum and identify a major structure contained in each. LECTURE LAB
9. **Peritoneal relationships.** Classify abdominopelvic organs as intraperitoneal or retroperitoneal using SADPUCKER and define the mesentery, omenta, and mesocolon. LECTURE

10. **Abdominopelvic regions and quadrants.** Locate organs on the nine-region and four-quadrant grids and name the planes that draw each grid. LECTURE LAB

CELL ANATOMY

11. **Generalized cell regions.** Name the three basic regions of a generalized cell, the plasma membrane, cytoplasm, and nucleus, and state what each contains. LECTURE

12. **Plasma membrane structure.** Describe the plasma membrane components, distinguish integral from peripheral proteins, and identify the surface specializations microvilli, cilia, and flagellum. LECTURE

13. **Nucleus structure.** Identify the nuclear envelope, nuclear pores, nucleolus, nucleoplasm, chromatin, and chromosomes and state the role of each. LECTURE

14. **Organelle identification.** Identify the membranous and non-membranous organelles and state the structural job of each, including the endomembrane system path from rough ER to exocytosis. LECTURE

TISSUES AND HISTOLOGY

15. **Embryonic germ layers.** Name the three primary germ layers, ectoderm, mesoderm, and endoderm, and the tissues each one gives rise to. LECTURE

16. **Epithelial tissue identification.** Classify epithelial tissue by cell layers and cell shape and identify the basement membrane, apical specializations, and glandular epithelium on a slide. LECTURE LAB

17. **Cell junctions.** Identify the five epithelial cell junctions, what each does, what it links to, and its major protein. LECTURE LAB

18. **Connective tissue identification.** Describe connective tissue as cells, fibers, and ground substance and recognize the connective tissue proper subtypes, cartilage, bone, and blood on a slide. LECTURE LAB

19. **Body membranes.** Distinguish the cutaneous, mucous, serous, and synovial membranes by composition and location. LECTURE

INTEGUMENTARY

20. **Skin layers and hypodermis.** Name the epidermis and dermis and the hypodermis beneath them and identify the tissue that composes each. LECTURE LAB

21. **Epidermal strata and cells.** List the five epidermal strata in order deep to superficial and identify the keratinocytes, melanocytes, tactile cells, and dendritic cells. LECTURE LAB

22. **Dermis layers.** Distinguish the papillary and reticular layers of the dermis by tissue and identify the structures and sensory corpuscles each contains. LECTURE LAB

23. **Accessory structures of the skin.** Identify the accessory structures of the skin including the parts of a hair and follicle, the parts of a nail, and the five gland types and their secretion modes. LECTURE LAB

Module 2. Skeletal System and Joints

30 COMPETENCIES · ASSESSED ON EXAM 2

BONE HISTOLOGY

24. **Cartilage types and structure.** Identify the three cartilage types by fiber content and location and label chondrocytes, lacunae, matrix, and perichondrium. LECTURE LAB

25. **Cartilage growth modes.** Distinguish appositional from interstitial cartilage growth by where each begins and the direction of expansion. LECTURE

26. **Bone classification by shape.** Classify a bone as long, short, flat, irregular, or sesamoid and give an example of each. LECTURE LAB

27. **Gross anatomy of a long bone.** Label the diaphysis, epiphysis, metaphysis, medullary cavity, articular cartilage, periosteum, endosteum, epiphyseal line, and marrow regions on a long bone. LECTURE LAB

28. **Compact and spongy bone microanatomy.** Label the parts of an osteon and contrast compact bone with the trabeculae of spongy bone by structure and location. LECTURE LAB

29. **The four bone cells.** Identify the osteogenic cell, osteoblast, osteocyte, and osteoclast and state the role and origin of each. LECTURE LAB

30. **Ossification and growth plate zones.** Contrast intramembranous and endochondral ossification and list the five growth plate zones in order from epiphysis to diaphysis. LECTURE

AXIAL SKELETON, SKULL

31. **Cranial and facial bones.** Distinguish the eight cranial bones from the fourteen facial bones and name the region each bone forms. LECTURE LAB

32. **Significant skull bone markings.** Identify the key markings on each cranial and facial bone, including features of the sphenoid, ethmoid, temporal, occipital, maxillae, and mandible. LECTURE LAB

33. **Sutures and fontanelles.** Locate the four major sutures and the pterion by the bones they join and name the four fontanelles and where each lies. LECTURE LAB

34. **Orbit, nasal cavity, palate, and sinuses.** Identify the bones forming the orbit, nasal septum, and hard palate and locate the four paranasal sinuses by the bone each occupies. LECTURE LAB

35. **Major skull foramina and contents.** Name the major skull foramina by the bone each pierces and state the nerve or vessel that passes through each. LECTURE LAB

AXIAL SKELETON, SPINE

36. **Vertebral column regions and curvatures.** Name the five regions of the vertebral column with the vertebra count of each and identify the four curvatures as primary or secondary. LECTURE LAB

37. **Typical vertebra and intervertebral disc.** Label the parts of a typical vertebra and the anulus fibrosus and nucleus pulposus of an intervertebral disc. LECTURE LAB

38. **Regional and specialized vertebrae.** Distinguish cervical, thoracic, and lumbar vertebrae by an identifying feature and describe the atlas, axis, dens, sacrum, and coccyx. LECTURE LAB

39. **Sternum and ribs.** Identify the parts of the sternum and rib markings and classify the twelve rib pairs as true, false, or floating by anterior attachment. LECTURE LAB

APPENDICULAR, UPPER EXTREMITY

40. **Pectoral girdle and its markings.** Identify the markings of the clavicle and scapula and explain how the clavicle links the upper limb to the axial skeleton. LECTURE LAB

41. **Humerus, radius, and ulna markings.** Identify the key markings of the humerus, radius, and ulna and state which forearm bone is lateral and which forms the elbow hinge. LECTURE LAB

42. **Bones of the hand.** Name the eight carpals in their two rows and the metacarpals and phalanges and locate the carpal tunnel and its contents. LECTURE LAB

APPENDICULAR SKELETON

43. **Bone markings vocabulary.** Classify a named bone marking as a projection, depression, or opening and match each term such as process, tuberosity, fossa, or sulcus to its description. LECTURE LAB

APPENDICULAR, LOWER EXTREMITY

44. **Hip bone and pelvis.** Name the three bones that fuse at the acetabulum, identify the markings of the ilium, ischium, and pubis, and describe the pelvis and its sex differences. LECTURE LAB

45. **Femur and patella markings.** Identify the key markings of the femur and the base and apex of the patella. LECTURE LAB

46. **Tibia and fibula markings.** Identify the markings of the tibia and fibula and state which bone bears weight and how to tell them apart. LECTURE LAB

47. **Bones and arches of the foot.** Name the seven tarsals, the metatarsals, and the phalanges and identify the three arches of the foot. LECTURE LAB

ARTICULATIONS AND JOINTS

48. **Structural and functional joint classification.** Classify a joint by structure as fibrous, cartilaginous, or synovial and by function as synarthrosis, amphiarthrosis, or diarthrosis and state the criteria for each. LECTURE LAB

49. **Fibrous and cartilaginous joints.** Describe the three fibrous joint types and the two cartilaginous joint types by structure, mobility, and example. LECTURE LAB

50. **Synovial joint structure and accessory parts.** Identify the structures of a synovial joint, its accessory structures such as menisci, labra, bursae, and tendon sheaths, and its nerve and blood supply. LECTURE LAB

51. **Movements at synovial joints.** Demonstrate and name the gliding, angular, rotational, and special movements permitted at synovial joints. LECTURE LAB

52. **Six synovial joint types.** Name the six synovial joint types with their axes of movement and give a body example of each. LECTURE LAB

53. **Shoulder and knee joint structure.** Describe the articulating bones and supporting ligaments of the glenohumeral and tibiofemoral joints and identify the rotator cuff, menisci, and cruciate and collateral ligaments. LECTURE LAB

Module 3. Cardiovascular, Muscle and Blood

42 COMPETENCIES · ASSESSED ON EXAM 3

TISSUES AND HISTOLOGY

54. **Muscle tissue identification.** Recognize the three muscle tissue types, skeletal, cardiac, and smooth, by striations, cell shape, and number of nuclei. LECTURE LAB

CARDIOVASCULAR

55. **Heart wall layers, pericardium, and internal features.** Name the three heart wall layers and the pericardial membranes, and locate internal features such as auricles, pectinate muscles, fossa ovalis, and trabeculae carneae. LECTURE LAB

56. **Heart chambers and septa.** Identify the four heart chambers and the interatrial and interventricular septa, and state what blood each chamber receives and where it sends it. LECTURE LAB

57. **Heart valves and what each separates.** Identify the four heart valves by type and location, and state which two chambers or vessels each one separates and where it prevents backflow. LECTURE LAB

58. **Chordae tendineae and papillary muscles.** Identify the chordae tendineae and papillary muscles and describe how they anchor and hold the atrioventricular valves shut. LECTURE LAB

59. **Cardiac muscle tissue and the intercalated disc.** Describe cardiac muscle tissue and identify the cardiomyocyte, striations, and intercalated disc with its desmosomes and gap junctions. LECTURE LAB

60. **Pathway of blood through the heart.** Trace one drop of blood through the four chambers and four valves of the heart in correct order. LECTURE LAB

61. **Great vessels and aortic arch.** Identify the great vessels attached to the base of the heart and state which circuit each one serves. LECTURE LAB

62. **Coronary circulation.** Trace coronary circulation from the aorta through the coronary arteries, myocardial capillaries, cardiac veins, and coronary sinus to the right atrium, and identify the major coronary vessels. LECTURE LAB

63. **Conduction system components and pathway.** Locate each component of the conduction system and trace the pathway in order from the SA node to the Purkinje fibers. LECTURE LAB

64. **Nerve supply to the heart.** Describe the autonomic nerve supply reaching the heart through the cardiac plexus and name the sympathetic and vagal parasympathetic sources. LECTURE

65. **Three tunics of a vessel wall.** Name the three tunics of a vessel wall, state the tissue each contains, and explain which layer forms a capillary wall. LECTURE LAB

66. **The five vessel types.** Compare arteries, arterioles, capillaries, venules, and veins by wall structure, direction of flow, and role. LECTURE LAB

67. **Kinds of artery, capillary, and vein.** Distinguish elastic and muscular arteries, continuous, fenestrated, and sinusoid capillaries, and identify venous valves and their supporting features. LECTURE LAB

68. **Circulatory routes, portal systems, and anastomoses.** Describe the pulmonary and systemic circuits and identify portal systems and anastomoses as alternate circulatory arrangements. LECTURE

69. **Vessel wall disorders, arterial and venous.** Describe atherosclerosis and how it changes an artery wall, and name common arterial and venous disorders by the structure each affects. LECTURE

70. **Fetal circulation, shunts, and adult remnants.** Identify the fetal vessels and three shunts, state how each reroutes blood, and name the adult remnant each becomes. LECTURE LAB

MUSCLE STRUCTURE

71. **Connective tissue coverings.** Name the epimysium, perimysium, and endomysium and trace how the three sheaths merge into the tendon that anchors muscle to bone. LECTURE LAB

72. **Levels of organization.** Order the nested levels of skeletal muscle from whole muscle through fascicle, fiber, and myofibril down to the myofilament. LECTURE

73. **Muscle fiber internal structure.** Identify the sarcolemma, sarcoplasm, myonuclei, myofibrils, sarcoplasmic reticulum, terminal cisternae, T tubules, and triad of a muscle fiber. LECTURE LAB

74. **Sarcomere bands and filaments.** Diagram a sarcomere and name the Z disc, A band, I band, H zone, M line, and zone of overlap, stating which filaments occupy each region. LECTURE LAB

75. **Myofilament and structural proteins.** Distinguish thick from thin filaments by protein and anchor point and identify the roles of titin, nebulin, alpha-actinin, myomesin, and dystrophin. LECTURE

76. **Three muscle tissue types.** Compare skeletal, cardiac, and smooth muscle by location, striations, control, and cell features including intercalated discs. LECTURE LAB

FASCICLE ARRANGEMENT

77. **Fascicle arrangement patterns.** Identify parallel, fusiform, circular, convergent, and pennate fascicle patterns and give an example muscle for each, relating architecture to power versus range of motion. LECTURE

78. **Muscle roles and naming.** Classify agonist, antagonist, synergist, and fixator roles in a movement and decode a muscle name from its direction, size, shape, location, action, origins, or attachments. LECTURE

79. **Lever systems.** Identify the fulcrum, effort, and load of a lever and classify first-, second-, and third-class levers with a body example of each. LECTURE

BLOOD

80. **Blood composition and plasma.** Describe blood as a fluid connective tissue and identify plasma, formed elements, hematocrit, buffy coat, and the plasma proteins albumin, globulins, and fibrinogen. LECTURE LAB

81. **Formed elements and leukocytes.** Identify erythrocytes, platelets, and the five leukocytes on a smear and describe erythrocyte structure and the granulocyte and agranulocyte groups. LECTURE LAB

82. **Hematopoiesis.** Explain where blood cells form and identify the hemocytoblast, megakaryocyte, and the myeloid and lymphoid lines of hematopoiesis. LECTURE LAB

83. **Blood disorders.** Name anemia, sickle cell disease, polycythemia, leukemia, leukopenia, thrombocytopenia, and hemophilia and identify the component each one affects. LECTURE

LAB: UPPER EXTREMITY MUSCLES

84. **Chest and anterior arm muscles.** Identify on the cadaver or model the pectoralis major, pectoralis minor, serratus anterior, deltoid, biceps brachii, brachialis, and coracobrachialis with their actions. LAB

85. **Forearm compartments.** Identify the anterior flexor and posterior extensor forearm muscles including flexor carpi radialis, palmaris longus, pronator teres, and extensor digitorum and state each compartment action. LAB

86. **Posterior shoulder and rotator cuff.** Identify the trapezius, latissimus dorsi, rhomboids, levator scapulae, and the four rotator cuff muscles supraspinatus, infraspinatus, teres minor, and subscapularis with their actions. LAB

LAB: UPPER EXTREMITY VESSELS

87. **Upper-extremity arteries and veins.** Trace and identify the subclavian, axillary, brachial, radial, and ulnar arteries and the cephalic, basilic, and median cubital veins on the upper limb. LAB

LAB: UPPER EXTREMITY NERVES

88. **Upper-extremity nerves.** Identify the major nerves of the upper limb including the musculocutaneous, median, ulnar, radial, and axillary nerves and the brachial plexus they arise from. LAB

LAB: MUSCLES

89. **Abdominal and pelvic wall muscles.** Identify the abdominal and pelvic wall muscles on the cadaver including the rectus abdominis, external and internal obliques, transversus abdominis, and the pelvic floor muscles. LAB

90. **Gluteal and thigh muscles.** Identify the gluteal and thigh muscles including the gluteal group, quadriceps, hamstrings, and adductor group on the cadaver. LAB

91. **Leg muscles.** Identify the muscles of the leg including the anterior, lateral, and posterior compartment muscles on the cadaver. LAB

HEAD AND NECK LAB MUSCLES

92. **Muscles of facial expression.** Identify the muscles of facial expression including occipitofrontalis, orbicularis oculi, orbicularis oris, buccinator, zygomaticus, and platysma on a specimen or model. LAB

93. **Muscles of mastication.** Identify the muscles of mastication masseter, temporalis, and the medial and lateral pterygoids on a specimen or model. LAB

94. **Muscles of the neck.** Identify the sternocleidomastoid, the scalenes, and the suprahyoid and infrahyoid muscles of the neck on a specimen or model. LECTURE LAB

95. **Extraocular muscles.** Identify the four rectus and two oblique extraocular muscles and the levator palpebrae superioris on a specimen or model. LECTURE LAB

Module 4. Respiratory, Lymphatic, Digestive and Endocrine

36 COMPETENCIES · ASSESSED ON EXAM 4

RESPIRATORY

96. **Upper and lower tracts, conducting and respiratory zones.** Distinguish the upper and lower respiratory tracts and the conducting and respiratory zones, and assign the major airway structures to each. LECTURE LAB

97. **Nose, nasal cavity, and pharynx.** Identify the parts of the nasal cavity, paranasal sinuses, and the three regions of the pharynx. LECTURE LAB

98. **Larynx.** Identify the cartilages and folds of the larynx, including the thyroid and cricoid cartilages, epiglottis, arytenoids, vocal cords, and glottis. LECTURE LAB

99. **Trachea and bronchial tree.** Trace air through the trachea and bronchial tree in order from the trachea and carina down to the terminal bronchioles. LECTURE LAB

100. **Lungs and pleurae.** Identify the lobes, fissures, and surfaces of each lung and the parietal and visceral pleurae with the pleural cavity. LECTURE LAB

101. **Respiratory zone and alveolar structure.** Describe the structure of an alveolus and identify the type I and type II alveolar cells, alveolar macrophages, respiratory membrane, and pulmonary capillaries. LECTURE LAB

102. **Thoracic wall and diaphragm.** Identify the diaphragm, the intercostal muscles, and the three structures passing through the diaphragm at T8, T10, and T12. LECTURE LAB

103. **Respiratory disorders by structure affected.** Name common respiratory disorders and state which airway, alveolar, or pleural structure each one affects. LECTURE

LYMPHATIC SYSTEM

104. Lymph pathway. Define lymph and trace its one-way route from lymphatic capillaries through collecting vessels, nodes, trunks, and ducts back to the subclavian veins. LECTURE

105. Vessel and node structure. Describe the structure of lymphatic capillaries, valved collecting vessels, and a lymph node including capsule, cortex, medulla, hilum, and afferent and efferent vessels. LECTURE LAB

106. Primary and secondary organs. Sort red bone marrow, thymus, lymph nodes, spleen, tonsils, and MALT into primary and secondary lymphoid organs with location and role. LECTURE

107. Lymphatic disorders. Name edema, lymphedema, lymphadenopathy, lymphoma, splenomegaly, and tonsillitis and identify the structure each one affects. LECTURE

ALIMENTARY CANAL

108. Four-layer gut wall. Name the four layers of the alimentary canal wall from lumen outward and state the dominant tissue of each. LECTURE LAB

109. Enteric nerve plexuses. Locate the submucosal and myenteric plexuses in the gut wall and match each to the wall layer it sits within. LECTURE LAB

110. Peritoneum and folds. Identify the parietal and visceral peritoneum, the peritoneal cavity, and the five folds greater omentum, lesser omentum, falciform ligament, mesentery, and mesocolon and state what each fold connects. LECTURE LAB

111. Retroperitoneal organs. Distinguish retroperitoneal organs from intraperitoneal ones and name the kidneys, pancreas, duodenum, and parts of the colon as retroperitoneal. LECTURE LAB

112. Mouth, pharynx, esophagus. Identify the oral cavity boundaries, hard and soft palate, uvula, fauces, tonsils, the three pharyngeal regions, and the esophagus with its hiatus. LECTURE LAB

113. Major canal sphincters. Locate the upper and lower esophageal, pyloric, ileocecal, and internal and external anal sphincters and state which regions each one separates. LECTURE LAB

114. Stomach regions and features. Identify the cardia, fundus, body, pyloric part, curvatures, rugae, and the three-layered muscularis of the stomach. LECTURE LAB

115. Small intestine and surface area. Identify the duodenum, jejunum, and ileum and name the circular folds, villi, lacteal, and microvilli as features that expand absorptive surface area. LECTURE LAB

116. Large intestine features. Identify the cecum, appendix, four parts of the colon, rectum, and anal canal along with the teniae coli, haustra, and omental appendices. LECTURE LAB

117. Trace the pathway of food. Trace a bite of food through every named segment of the alimentary canal in order from mouth to anus. LECTURE LAB

118. Clinical anatomy of canal disorders. Match GERD, hiatal hernia, peptic ulcer, appendicitis, diverticulosis, IBD, hemorrhoids, and colorectal cancer to the canal structure each one affects. LECTURE

ACCESSORY ORGANS

119. Teeth and tongue. Identify the teeth and tongue with the lingual frenulum, papillae, intrinsic and extrinsic muscles, and lingual glands. LECTURE LAB

120. Salivary glands. Locate the parotid, submandibular, and sublingual glands and state where the duct of each opens. LECTURE LAB

121. Liver gross and lobule anatomy. Identify the four liver lobes, falciform ligament, and ligamentum teres, and name the hepatocytes, sinusoids, canaliculi, and portal triad of the lobule. LECTURE LAB

122. Gallbladder and bile pathway. Identify the gallbladder regions and trace bile from the canaliculi through the hepatic, cystic, and common bile ducts to the hepatopancreatic ampulla and duodenum. LECTURE LAB

123. Pancreas structure. Identify the head, body, and tail of the pancreas with its main and accessory ducts, major duodenal papilla, sphincter of Oddi, acini, and islets. LECTURE LAB

124. Clinical anatomy of accessory disorders. Match gallstones, cholecystitis, hepatitis, cirrhosis, jaundice, pancreatitis, and mumps to the accessory organ each one affects and explain how a stone at the ampulla can block bile and pancreatic flow at once. LECTURE

LAB: BV/N 3

125. Abdominal aorta and portal system. Identify the major branches of the abdominal aorta and the vessels of the hepatic portal system on the cadaver. LAB

126. Lower limb vessels and nerves. Identify the femoral and lower-limb blood vessels and the major nerves of the abdomen and lower limb on the cadaver. LAB

ENDOCRINE

127. Major endocrine glands and locations. Name the major endocrine glands and locate each one in the body from the hypothalamus to the gonads. LECTURE

128. **Pituitary gross anatomy and lobes.** Describe the pituitary in the sella turcica, its infundibulum link to the hypothalamus, and compare the anterior and posterior lobes by tissue type. LECTURE LAB

129. **Thyroid and parathyroid anatomy.** Identify the thyroid lobes, isthmus, follicles, colloid, follicular and parafollicular cells, and the parathyroid chief and oxyphil cells. LECTURE LAB

130. **Adrenal gland zones.** Identify the adrenal cortex zones in order and the adrenal medulla with its chromaffin cells. LECTURE LAB

131. **Pancreas, gonads, and organs with endocrine tissue.** Locate the pancreatic islets and gonads and name other organs such as the heart, kidneys, and skin that contain endocrine tissue. LECTURE

Module 5. Urinary, Reproductive and the Nervous System

62 COMPETENCIES · ASSESSED ON EXAM 5

TISSUES AND HISTOLOGY

132. **Nervous tissue identification.** Identify neurons and neuroglia in nervous tissue and the parts of a multipolar neuron. LECTURE LAB

URINARY

133. **Kidney gross anatomy and layers.** Describe the location and tissue layers of the kidney and identify the hilum, capsule, adipose capsule, and renal fascia. LECTURE LAB

134. **Internal kidney regions.** Identify the cortex, medulla, renal columns, pyramids, papillae, parenchyma, calyces, and renal pelvis on a sectioned kidney. LECTURE LAB

135. **Nephron structure.** Identify the renal corpuscle, glomerulus, glomerular capsule, podocytes, PCT, nephron loop, DCT, and collecting duct of a nephron. LECTURE LAB

136. **Cortical vs juxtamedullary nephrons.** Compare cortical and juxtamedullary nephrons by loop depth and associated capillary beds. LECTURE LAB

137. **JGA and filtration membrane.** Identify the juxtaglomerular apparatus, macula densa, and JG cells, and name the three layers of the filtration membrane and what each holds back. LECTURE LAB

138. **Kidney blood supply pathway.** Trace blood through the kidney from the renal artery to the renal vein, naming each vessel in order. LECTURE LAB

139. **Ureters, bladder, urethra and urine path.** Trace urine from the collecting duct to the outside and identify the ureter, bladder wall layers, trigone, detrusor, sphincters, and male and female urethra. LECTURE LAB

140. **Clinical anatomy of urinary disorders.** Relate urinary disorders such as nephroptosis, kidney stones, and UTI to the anatomical structure or pathway each one affects. LECTURE

MALE REPRODUCTIVE

141. **Scrotum and testis structure.** Identify the scrotum, dartos and cremaster muscles, tunica layers, seminiferous tubules, sustentacular and interstitial cells, rete testis, and efferent ductules. LECTURE LAB

142. **Sperm duct system and pathway.** Trace the path of sperm from the seminiferous tubules to the outside, naming the epididymis, ductus deferens, ejaculatory duct, urethra, and spermatic cord contents. LECTURE LAB

143. **Sperm cell structure.** Identify the head, acrosome, neck, middle piece, principal piece, and end piece of a sperm cell. LECTURE LAB

144. **Accessory glands and penis.** Name the seminal vesicles, prostate, and bulbourethral glands and identify the root, body, glans, prepuce, and three erectile columns of the penis. LECTURE LAB

145. **Clinical anatomy of male disorders.** Relate male reproductive disorders such as cryptorchidism and benign prostatic hyperplasia to the affected structure and its position. LECTURE

FEMALE REPRODUCTIVE

146. **Ovary structure and follicle stages.** Identify the ovary layers, ligaments, and follicle stages including the corpus luteum and corpus albicans. LECTURE LAB

147. **Uterine tubes and ovum pathway.** Trace the oocyte from the ovary to the uterus, naming the fimbriae, infundibulum, ampulla, and isthmus. LECTURE LAB

148. **Uterus regions, ligaments, and wall.** Name the regions and supporting ligaments of the uterus and identify the perimetrium, myometrium, and endometrium. LECTURE LAB

149. **Vagina and external genitalia.** Identify the vagina, fornix, hymen, and the vulvar structures including the clitoris, vestibule, labia, and greater vestibular glands. LECTURE LAB

150. **Mammary gland structure.** Identify the lobes, lobules, alveoli, lactiferous ducts, nipple, areola, and suspensory ligaments of the mammary gland. LECTURE LAB

BRAIN

151. **Brain gross organization.** Identify the four major brain divisions cerebrum, diencephalon, brainstem, and cerebellum, and distinguish the cortical gray matter from the deep white matter on a specimen. LECTURE LAB

152. **Neural tube brain development.** Trace brain development from the neural tube through the primary and secondary vesicles and match each secondary vesicle to its adult structures and ventricle. LECTURE

153. **Cerebral surface and lobes.** Identify the cerebral hemispheres, five lobes, gyri, sulci, and the named central, lateral, and parieto-occipital sulci along with the longitudinal and transverse fissures on a specimen. LECTURE LAB

154. **Functional cortical areas.** Map the motor, sensory, and association areas of the cortex to their gyri and lobes, including primary motor, primary somatosensory, primary visual, auditory, gustatory, and olfactory cortices, Broca and Wernicke areas, and the prefrontal cortex. LECTURE LAB

155. **Cerebral white matter tracts.** Compare association, commissural, and projection fibers by what each connects and identify the corpus callosum and internal capsule on a specimen. LECTURE LAB

156. **Basal ganglia.** Identify the caudate nucleus, putamen, and globus pallidus and locate the functionally grouped substantia nigra and subthalamic nucleus. LECTURE LAB

157. **Diencephalon.** Identify the thalamus with the massa intermedia, the hypothalamus, the epithalamus with the pineal gland, and the third ventricle they enclose. LECTURE LAB

158. **Cerebellum.** Identify the cerebellar hemispheres, vermis, folia, arbor vitae, deep nuclei, and the three cerebellar peduncles with their brainstem connections. LECTURE LAB

159. **Limbic system.** Locate the cingulate gyrus, amygdala, and hippocampus as the limbic structures ringing the diencephalon. LECTURE LAB

160. **Cerebral arterial supply.** Trace the anterior and posterior circulations through the Circle of Willis and identify the anterior, middle, and posterior cerebral arteries, the communicating arteries, the internal carotid, vertebral, and basilar arteries, and the cerebellar branches. LECTURE LAB

BRAINSTEM

161. **Brainstem regions.** Locate the midbrain, pons, and medulla oblongata in order and describe how the brainstem connects the higher brain to the spinal cord. LECTURE LAB

162. **Midbrain structures.** Identify the tectum with superior and inferior colliculi, the cerebral peduncles, the substantia nigra, and the cerebral aqueduct on a midbrain specimen. LECTURE LAB

163. **Pons structures.** Identify the pons, its pontine nuclei, the middle cerebellar peduncles, and the fourth ventricle behind it. LECTURE LAB

164. **Medulla oblongata structures.** Identify the pyramids, olives, and the decussation of the pyramids on the medulla and locate the vital cardiac, respiratory, and vasomotor centers. LECTURE LAB

165. **Cranial nerve nuclei distribution.** Match cranial nerve nuclei III through XII to the brainstem region that houses them. LECTURE LAB

166. **Reticular formation.** Locate the reticular formation as a net of gray matter through the brainstem core and distinguish its ascending reticular activating system from its descending component. LECTURE

MENINGES AND CSF

167. **Cranial meninges and dural reflections.** Identify the dura, arachnoid, and pia mater and the dural reflections falx cerebri, falx cerebelli, and tentorium cerebelli, and trace dural venous sinus drainage from the superior sagittal sinus to the internal jugular vein. LECTURE LAB

168. **Spinal meninges and meningeal spaces.** Identify the spinal meninges, the denticulate ligaments and filum terminale, and distinguish the epidural, subdural, and subarachnoid spaces by location and contents. LECTURE LAB

169. **Cord termination and lumbar puncture.** Identify the conus medullaris, cauda equina, and lumbar cistern and list in order the layers a lumbar puncture needle crosses from skin to subarachnoid space. LECTURE LAB

170. **Ventricular system.** Identify the lateral, third, and fourth ventricles and the central canal and describe how the interventricular foramina, cerebral aqueduct, and fourth ventricle apertures connect them. LECTURE LAB

171. **CSF production and circulation.** Trace CSF from production at the choroid plexus through the ventricles and subarachnoid space to reabsorption at the arachnoid granulations. LECTURE LAB

172. **Blood-brain barrier structure.** Name the three structural components of the blood-brain barrier the tight-junctioned capillary endothelium, the basement membrane, and the astrocyte end-feet. LECTURE LAB

FUNCTIONAL ORGANIZATION AND NERVOUS TISSUE

173. **Nervous system divisions.** Outline the organization of the nervous system into CNS and PNS and its somatic, autonomic, and enteric motor divisions. LECTURE

174. **Neuron structure.** Identify the cell body, dendrites, axon, axon hillock, Nissl bodies, and axon terminals of a neuron and state the direction a signal travels through them. LECTURE LAB

175. **Synapse structures.** Identify the components of a chemical synapse including synaptic end bulbs, synaptic vesicles, and synaptic cleft, and name the three synapse types by the site they contact. LECTURE LAB

176. **Neuron classification.** Classify neurons structurally as multipolar, bipolar, or unipolar and functionally as sensory, motor, or interneuron with an example of each. LECTURE LAB

177. **Neuroglia identification.** Name the four CNS glia and two PNS glia and describe the support role each one performs. LECTURE LAB

178. **Myelination and regeneration.** Describe the myelin sheath, nodes of Ranvier, and neurolemma and contrast PNS and CNS myelination by the glial cell involved and by regeneration potential. LECTURE LAB

GROSS ANATOMY AND NEURONAL INTEGRATION

179. **CNS and PNS tissue terms.** Define nucleus, ganglion, tract, and nerve and state whether each names a cluster of cell bodies or a bundle of axons in the CNS or PNS. LECTURE

180. **Gray and white matter.** Distinguish gray matter from white matter by composition and locate each in the brain and the spinal cord. LECTURE LAB

181. **Reflex arc components.** Name the five components of a reflex arc in order from receptor to effector and trace a signal through them. LECTURE

182. **Types of reflexes.** Compare somatic versus autonomic, monosynaptic versus polysynaptic, and innate versus learned reflexes and give an example of each. LECTURE

183. **Neuronal pools and pathways.** Describe convergence and divergence circuits and distinguish ascending pathways, descending pathways, and decussation. LECTURE

THE SPINAL CORD

184. **Spinal cord external anatomy.** Identify the cervical and lumbar enlargements, conus medullaris, cauda equina, filum terminale, anterior median fissure, and posterior median sulcus. LECTURE LAB

185. **Roots, rootlets, and root ganglion.** Identify the rootlets, posterior and anterior roots, posterior root ganglion, and intervertebral foramen and state which root carries sensory versus motor axons. LECTURE LAB

186. **Spinal cord internal anatomy.** Identify the gray horns, central canal, commissures, and white columns on a cord cross-section and state the contents of the posterior, lateral, and anterior horns. LAB

THE PERIPHERAL NERVOUS SYSTEM

187. **Nerve connective tissue wrappings.** Identify the axon, fascicle, endoneurium, perineurium, and epineurium of a peripheral nerve and state the level each layer wraps. LECTURE LAB

188. **Spinal nerves and rami.** State the count of the 31 spinal nerve pairs by region and identify the posterior ramus, anterior ramus, meningeal branch, and rami communicantes by the territory each supplies. LECTURE LAB

189. **Sensory receptor classification.** Classify sensory receptors by stimulus, location, and structure and match named receptors such as tactile corpuscles, lamellar corpuscles, and muscle spindles to their location and stimulus. LECTURE LAB

THE CRANIAL NERVES

190. **Cranial nerves.** Name the twelve cranial nerves in order, classify each as sensory, motor, or mixed, and match each to its primary function and skull foramen. LECTURE LAB

THE NERVE PLEXUSES

191. **Nerve plexuses.** Name the cervical, brachial, lumbar, sacral, and coccygeal plexuses with the spinal nerves that form each and predict the body region affected if a given plexus is damaged. LECTURE LAB

THE AUTONOMIC NERVOUS SYSTEM

192. **Autonomic two-neuron pathway.** Contrast the somatic and autonomic motor systems and trace the two-neuron pathway through a preganglionic neuron, autonomic ganglion, and postganglionic neuron to an effector. LECTURE

193. **Autonomic ganglia and divisions.** Compare the sympathetic and parasympathetic divisions by CNS origin and ganglion location and identify the sympathetic trunk, prevertebral ganglia, terminal ganglia, and autonomic and enteric plexuses. LECTURE